Tait Weicht

602-684-7699
tjweicht@ucdavis.edu
www.linkedin.com/in/tait-weicht
https://github.com/lemniscate8
https://taitweicht.com

Skills and Qualifications

- Self-driven exploration in diverse subjects, quick mastery of new mathematics at a graduate level
- Proficiency in Python, C++, JavaScript, TypeScript, Java; some experience with C#, Prolog, SQL
- Experience manipulating data tables in R and using *dplyr*'s data manipulation verbs
- Experience with Git version control, React (with hooks), the styled components ecosystem, and Bootstrap
- Desire to collaborate and produce quality work with other motivated co-workers

Education

University of California, Davis
PhD in Applied Mathematics — Expected September 2026

Seattle Pacific University, Seattle, WA
BS in Computer Science, *summa cum laude* — Spring 2020
BS in Mathematics, *summa cum laude* — Spring 2020

Experience

Researcher, advised by Alexander Wein, *UCD* — June 2023 - Ongoing
- Developed methods for computer-aided proof of sharp bounds on an algorithm for tensor decomposition
- Devised method for orbit recovery in multi-reference alignment variants with variety-constrained linear systems (papers: *https://arxiv.org/abs/2504.17947*, *https://arxiv.org/abs/2606.11701*)

Researcher, advised by Albert Fannjiang, *UCD* — June 2022 - Ongoing
- Exploring inverse problems related to imaging and geometric tomography for UCD4IDS
- Determining the effect of dose-fractionation on spectral method for phase retrieval in X-ray diffraction using the non-asymptotic random matrix theory for spiked Wishart distributions (manuscript in progress)

Research Software Developer - Python — September 2024 - June 2026
University of California, Davis
- Implemented a robust version of the simultaneous diagonalization algorithm for tensor decomposition
- Implemented a heretofore theoretical algorithm computing the intersection of a linear subspace with a conic variety (see *https://github.com/lemniscate8/linear-intersect-varieties*)
- Applied these algorithms to achieve orbit recovery in dihedral multi-reference alignment (see *arXiv* paper)
- Use unit testing to validate results

Frontend Developer - C++ — June - August 2021
Michigan State University - WAVES Program
- Created new graphical user interfaces for C++ driven web applications in the header-only library, Empirical
- Suggested a new schema for component inheritance to ensure that prefabricated web components are extensible
- Presented at 2021 BEACON conference, with proof-of-concept using C++ code live in the slides

Frontend Development Intern - React, HTML, C# — June - August 2019
Faithlife Corporation, Bellingham, WA
- Created front-end components in React for a communications web application faithful to user interface and experience design specifications
- Wrote unit tests for backend functions in C# to store and retrieve user data from a SQL database

Technical Writer/Researcher — February 2019
COMAP's Interdisciplinary Contest in Modeling
- Worked in a team of 3 to write and submit a solution paper in 5 days (Problem D, Team #1908810)
- Leveraged spectral graph theory and agent-based modeling to model flow on a network
- Identified potential bottlenecks in the Louvre's floor-plan and gave recommendations for an evacuation strategy